NOTICE OF INTENT

This document provides information concerning the major tasks and typical responsibilities associated with a 6001B Major Inspection. It describes the overall work scope, but does not detail specific methods. Its primary purpose is to provide an overview of a typical inspection process. Design changes subsequent to the issue date of this document may not be reflected in this guide.

This document does not purport to cover all aspects of a particular inspection. Unit-specific details are provided in the document and drawing packages supplied with each requisition, and all sketches and descriptions are to be considered "Typical." This document is property of General Electric and is considered Company Proprietary Information.

HEALTH AND SAFETY REQUIREMENTS

The GI&FS Environmental Health & Safety Manual (EHSM) outlines the requirements that must be followed on the job site during an installation or maintenance outage.

Safety, health, and protection of the environment must be an integral part of all business operations, design, construction and start-up, and includes EHS planning prior to and during work to anticipate and minimize hazards and incidents. Every reasonable effort must be made to create and maintain a safe and healthy work environment at each customer site.

Where working conditions are not covered by these safety requirements and procedures, or when a requirement or a procedure is not completely understood, obtain specific instructions or clarification from manager, supervisor or EHS before proceeding with the work.

INTRODUCTION

This Maintenance Procedure currently exists as a part of a Gas Turbine Service Manual, and is intended to establish a consistent approach to the performance of flange-to-flange maintenance inspections for a MS6001B gas turbine. This procedure is intended to consolidate actual field experience and other maintenance related information normally available only from widely diverse sources. As maintenance experience is gained on this unit, this procedure will be supplemented with additional information and descriptions of the work to be performed. This article currently provides inspection information for Major Inspection workscope. The procedures described in the text do not supersede or replace specific product department documentation.

REFERENCE DOCUMENTS

No.	Description
EM 5202	Gas Turbine Mechanical Support and Jacking Instructions
MLI 0404	Clearance Diagram

MAJOR INSPECTION

Caution **Before proceeding with a major inspection ensure that the gas turbine electrical power is tagged-out; CO2 system is deactivated, liquid fuel system is purged, deactivated and/or the gas supply is blanked off.**

DISASSEMBLY

OPERATION 1 - PERFORM ALIGNMENT CHECKS PRIOR TO UNIT DISASSEMBLY

NOTE It is suggested that assistance be obtained from your GE Field Service Representative when performing alignment checks.

1. Remove the load coupling guard and screen.

2. Remove the accessory coupling guard bottom cover. Remove the bolting from the accessory gear end of the guard and the two top bolts which connect the guard to the turbine end coupling guard section. Lift this section off.

3. Remove the bolting from the turbine end of the coupling guard and lift off this section.

4. Bag all removed hardware and identify for reassembly.

5. Check turbine to load gear alignment.

   NOTE Use the hydraulic ratchet to rotate the unit for turbine to load gear alignment checks. Indicate both face and rim readings on the stationary load gear rotor.

   a. Check alignment of turbine to load gear at the load gear end of the load coupling. This allows an alignment procedure which is identical to the one employed during installation. The turbine to load gear coupling must be unbolted and jacked apart. Matchmark load coupling prior to unbolting and jacking apart.

      NOTE The output coupling is accessible for removal and replacement during major inspection since the exhaust frame is disassembled, however special fixturing (tooling) is provided to remove and replace the output coupling and to disassemble and assemble the number 2 bearing with the exhaust frame in place.



All upper half casings must be installed and the turbine supported on its own support mounts prior to checking alignment.

b. Position a dial indicator for a vertical reading on the O.D. of the load coupling every 45°. Rotate turbine rotor and record the opening coupling runout on GT5025. Refer to MLI 0411, Basic Field Alignment Instructions.



The generator field and load gear do not have a thrust bearing; therefore, there will be a tendency for the rotor to "walk" axially.

c. The "walking" can be compensated for by using two dial indicators. One indicator will be mounted on the rotating flange and will measure the actual face run-out. The other indicator will mount to the stationary flange and it will see mechanical face run-out plus axial movement, if any. Both indicators must be read simultaneously.



All readings must repeat three consecutive times to be considered accurate. The indicators should return to "0" (corrected reading of zero for the face and rim indicators) when returning to the top end of the vertical mark. If they do not, check tightness of set-up and repeat the readings. Refer to the Basic Alignment Instructions in the Service Manual.

d. Record coupling alignment readings on Inspection Form GT1010. Refer to MLI 0411, Basic Field Alignment Instructions.

6. Remove and inspect accessory coupling.

a. Matchmark the accessory gear to the coupling sleeve, both sleeves to the coupling shaft, and the turbine rotor to the coupling sleeve.

b. Set up a dial indicator and check end float and radial clearance.

c. Drain the oil, and unbolt the sleeves. Remove the coupling from the unit.

d. Clean the sleeves and teeth.

e. Record the radial clearance, end thrust, and tooth condition on Inspection Form GT7200.

7. Check turbine to accessory gear alignment.



The method described utilizes an alignment fixture supplied in the turbine tool kit, in place of the coupling shaft.

a. Using a felt tipped pen, mark two lines (one vertical, the other horizontal) across the face of the turbine coupling. The lines should intersect at the center of the coupling face.

b. Tightly attach the alignment fixture to the accessory gear coupling flange.

c. Place dial indicator to read from the alignment fixture to the face of the turbine coupling.

NOTE Using a piece of 2 x 4 mounted behind the accessory gear drive coupling flange, apply just enough force to take up the end float (thrust) on the shaft when taking the face readings.

d. The indicator must be placed to permit face readings to be taken without being affected by end float (thrust) of the accessory gear shaft when turning the shaft.

e. Rotate the accessory gear coupling with the alignment fixture and dial indicator attached until indicator is aligned on the top vertical face mark. Set dial indicator to read zero.

f. Rotate the accessory gear coupling (and alignment fixture) through one full revolution (360°) take readings at each face mark (90°). Record the dial indicator readings.

NOTE All readings must repeat three consecutive times to be considered accurate. The indicators should return to "0" (corrected reading of zero for the face and rim indicators) when returning to the top end of the vertical mark. If they do not, check tightness of set-up and repeat readings. Refer to the Basic Alignment Instructions in the Service Manual.

g. Place dial indicator to read from the fixture to the rim of the turbine coupling flange.

h. Take rim readings in the same manner, except that no correction is necessary for axial movement of the shaft. Record readings on Inspection Form GT1005.

OPERATION 2 - PERFORM COMBUSTION INSPECTION DISASSEMBLY OPS. 1 - 16

1. Prepare turbine compartment roof for removal.
2. Remove turbine compartment roof and side panels.
3. Remove liquid fuel lines.
4. Remove atomizing air lines.

NOTE In addition to the atomizing air lines to the fuel nozzles, unbolt and remove the upper half atomizing air manifold pipe also.

5. Remove gas fuel lines (if applicable).

NOTE In addition to the gas fuel lines to the fuel nozzles, unbolt and remove the upper half gas manifold pipe also.

6. Remove steam injection lines (if applicable).

7. Remove water injection lines (if applicable).

8. Remove liquid fuel check valves.

9. Remove false start drain tubing and manifold header.

10. Remove fuel nozzles.

11. Remove flame detectors and spark plugs.

12. Remove 11th stage cooling and sealing air lines.



The 5th stage cooling and sealing air lines and the bearing sealing air valve (VA14), if used, must also be removed from the upper half compressor casing at this time.

13. Unbolt and open up combustion outer casing covers.

14. Remove crossfire tube retainers, crossfire tubes, combustion liners and forward flow sleeves.

15. Remove atomizing air extraction piping manifold from turbine casing.

16. Remove outer combustion casings.

OPERATION 3 - REMOVE EXHAUST AND INLET DUCT ACCESS PANELS

1. On straight up exhaust configurations, unbolt the removable exhaust duct transition piece panel on the forward side of the panel.

2. On side exhaust configurations remove any architectural lagging, support members which may interfere with turbine casing removal. Remove the cowl section from the top of the exhaust plenum. This will facilitate access to the exhaust hood for removal.

3. Remove the 11th stage extraction pipe and compressor bleed valve by unbolting the bleed valve flange at compressor, loosen the "U" bracket, bellows clamp, and tubing connections and slip the piping out of the slip joint.

4. Remove the forward and aft flex seal clamp bars and the flex seal plates.

5. Unbolt the forward exhaust plenum removable access panel. Rig to lift the panel.

6. Install a length of 4 inch pipe or timber and a mechanical jack across the horizontal opening of the plenum and spread the panel sufficiently enough to remove the access panel. Lift the panel clear of the unit.

7. Unbolt and remove the inlet plenum access door located in the lower right hand section of the plenum, looking with airflow.

OPERATION 4 - TAKE INITIAL COMPRESSOR AND TURBINE ROTOR POSITIONING CHECKS

1. Refer to EM5321, Hot Gas Path Inspection, disassembly Operation number 3 in this instruction for positioning checks procedure.

OPERATION 5 - REMOVE INLET DUCT ELBOW

NOTE Due to variations in design of inlet duct work, refer to the Duct Arrangement, Inlet drawing (MLA041) for bolting details.

1. Support the elbow to prevent damage to the expansion joint when the elbow is released from the adjacent duct work.

2. Unbolt the duct work from the elbow and the transition piece.

Caution **The inlet duct elbow weighs approximately 7100 pounds and should be rigged accordingly when lifting.**

3. Lift the elbow straight up out of the lagging and set on blocking which supports the throat and aft gussets. Do not use the expansion joint for support.

4. Unbolt the transition piece from the inlet plenum.

Caution **The inlet transition piece weighs approximately 3400 pounds and should be rigged accordingly when lifting.**

5. Lift the transition piece from the inlet plenum and set on blocking. Cover the top opening.

OPERATION 6 - REMOVE THE UPPER HALF OF THE INLET PLENUM

1. Remove the inlet extension wings on the left and right sides of the plenum.

   NOTE The inlet plenum extension wings weigh approximately 1600 pounds and should be rigged accordingly for lifting.

2. Remove or disconnect any conduit runs, wiring, CO2 piping or any other items attached to the upper half of the forward and aft inlet plenum walls.

3. Unbolt the vertical and horizontal joints on the forward and aft walls.

4. Rig to lift the plenum using two lengths of timber spacing wedged forward to aft on the outer extremities to keep the plenum from disforming when being lifted.

   NOTE The upper half inlet plenum weighs approximately 1200 pounds and should be rigged accordingly for lifting.

5. Check to ensure the upper half plenum is mechanically free.

6. Lift the upper half plenum off and set on suitable blocking.

OPERATION 7 - PLACE MECHANICAL SUPPORT JACKS UNDER UNIT CASINGS AND INLET BELLMOUTH

1. Install mechanical support jacks per EM 5202.

OPERATION 8 - REMOVE UPPER HALF EXHAUST FRAME

NOTE The rail beam, trolley hoist and spreader bar are applicable to vertical exhaust stack configurations only. On side exhaust stack configurations the cowl cover on the exhaust plenum is removed for access to the exhaust frame for removal using an overhead crane and normal rigging.

1. Install the rail beam, trolley hoist, and spreader bar used for exhaust frame and turbine casing removal inside the exhaust duct transition piece.

2. Disconnect and remove as necessary all instrumentation tubing for wheelspace thermocouples and pressures inside the inner barrel tunnel.

3. Remove the left and right side outer horizontal joint insulating packs and identify for location.

4. Remove the outer insulating pack on the upper half aft vertical flange.

5. Remove the upper half vertical bolting between turbine case and exhaust frame.

6. Remove outer horizontal joint bolting left and right sides.

7. Grind off the welds from the bolt heads on the outer diffuser to inner diffuser strut ends (7 total) on top half only and remove the bolts.

8. Remove the inner tunnel vertical joint top half bolts between the exhaust frame and exhaust diffuser.

9. Remove the inner tunnel horizontal joint bolts left and right sides.

10. Move the trolley hoist over the center of gravity approximately 21 inches from the forward flange of the exhaust frame. Lower the spreader bar until it is about 8-10 inches from the top of the exhaust frame.

11. Rig by attaching one end of the wire rope sling to one of the lower lugs on the spreader bar. Feed the other end through the lifting lugs on the exhaust frame and attach to the lug on the other side of the spreader bar. Repeat for the other side.



The upper half exhaust frame assembly weighs approximately 4400 pounds.

12. Install the jacking bolts and jack the exhaust frame apart. Using the trolley hoist, gradually lift the exhaust frame assembly to the maximum height allowed by the hoist.

13. Move the trolley hoist out of the exhaust duct transition piece to the trolley hoist stop. Using an overhead crane, rig to the outer upper lugs of the spreader bar and transfer the load to the overhead crane. Lift the exhaust frame clear of the unit and set on suitable cribbing.

OPERATION 9 - REMOVE FORWARD COMPRESSOR CASING



Later vintage units have a combined (one piece) forward and aft compressor casing. Earlier units have a separate forward and aft compressor casing. The following operation pertains to the combined case or the forward case only, which ever is applicable. The aft compressor casing, if applicable, will be removed with the compressor discharge casing as a single unit when removed.

1. Remove the upper half bolting between the forward compressor and/or compressor casing, inlet and aft compressor and/or compressor discharge casing vertical flanges. Pull the two upper vertical joint dowels and tag for location.

2. Remove the horizontal flange bolting.

3. Remove the horizontal flange body bound bolts.

NOTE When removing the horizontal joint body bound bolts, they should be driven upwards.

4. Utilizing the jack bolts and guide pins provided, separate the casing at the horizontal joint, rig and lift the casing clear and set on suitable blocking. See Fig. MI-D1.

Caution **The forward compressor upper half casing weighs approximately 1775 pounds. The combined one piece upper half casing (forward and aft) weighs 3100 pounds.**

OPERATION 10 - REMOVE UPPER HALF INLET CASING

1. Disconnect the wiring to the magnetic pickup located in the inlet and number one bearing area.

2. Unbolt and remove the magnetic pickup ring from the front of the inlet and number one bearing.

3. Remove upper half inlet casing horizontal joint bolting.

Caution **There are twelve point bolts located in the air passage of the inlet casing, which must be removed. See Figure MI-D2.**

4. Remove the horizontal flange body bound bolts located on the forward inlet casing bellmouth.

5. Adjust the inlet guide vane control ring support plate bolts so that the lower half inlet guide vane control ring does not drop. See Fig. MI-D3.

6. Unbolt the inlet guide vane control ring and secure to the upper half casing.

Caution **Do not lift the inlet casing by using the inlet flange spreader bars. These are dampening struts and are not to be used for lifting attachments.**

7. Rig lifting cables, chainfall and blocks as shown on the Weights and Center of Gravity drawing in the Reference Drawing Section of this Service Manual to the upper half inlet casing.

Caution **The upper half inlet casing weighs approximately 5500 pounds.**

8. Use the chainfall to initially lift the inlet casing, continue to lift the casing straight up and clear of the unit. Use suitable blocking, stand the inlet casing in a vertical position.

9. Roll lower half rub ring around the shaft and remove.

NOTE The upper half rub ring is lifted as part of the upper half inlet casing.

OPERATION 11 - PERFORM HOT GAS PATH DISASSEMBLY OPERATIONS 6 - 11

NOTE Refer to EM5321, Hot Gas Path Inspection, for the following disassembly operations.

1. Remove turbine casing bolts and upper half first stage nozzle eccentric pin.

2. Remove upper half turbine casing.

3. Take turbine clearance checks.

4. Unbolt and remove transition pieces.

5. Take first stage nozzle radial concentricity checks.

6. Remove upper half first stage nozzle.

OPERATION 12 - REMOVE UPPER HALF FIRST STAGE NOZZLE SUPPORT RING

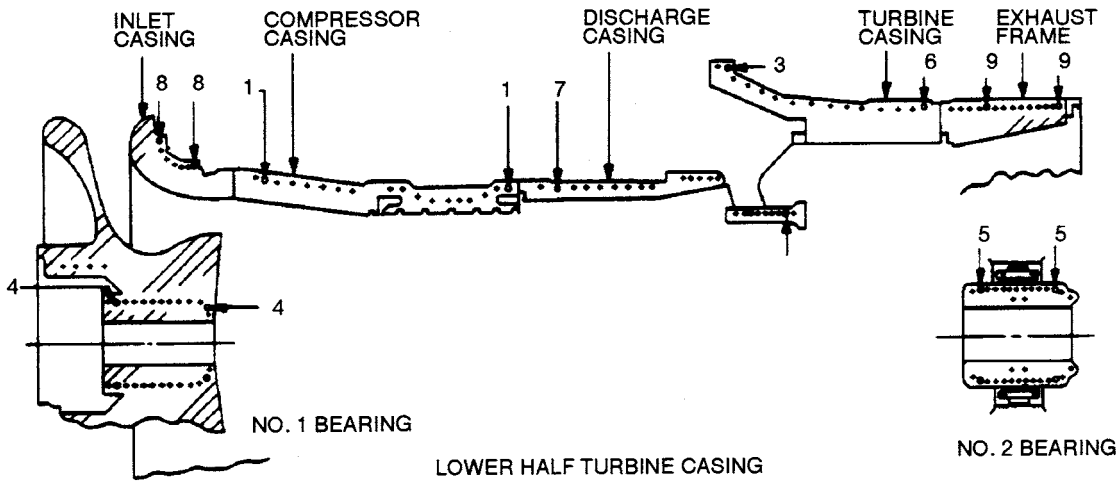
1. Unbolt and remove the support ring horizontal bolts.

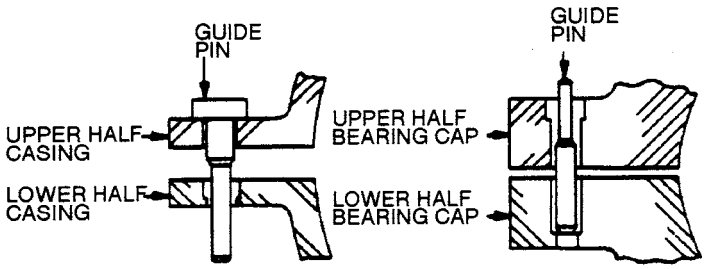
2. Unbolt the upper half vertical bolting from the compressor discharge case inner barrel.

3. Rig to the tapped holes provided and lift the support ring clear of the unit. Set on suitable blocking to protect from damage.

4. Remove and identify the support ring shim plates. See Fig. MI-D4.

5. Bag and identify bolting.





NOTES: GUIDE PINS FOUND IN TOOL KIT, MAJOR DISASSEMBLY

GUIDE PINS ARE USED IN BODY ROUND STUD HOLES OF CASINGS & DOWEL HOLES OF BEARING CAPS.

ITEM	GUIDE PIN	WHERE USED
1	248A4880 P4	COMPRESSOR CASING
3	295A9609 P2	TURBINE CASING
4	295A9601 P1	NO. 1 BEARING CAP
5	295A9601 P2	NO. 2 BEARING CAP
6	295A9609 P3	TURBINE CASING
7	295A9609 P1	DISCHARGE CASING
8	295A9600 P1	INLET CASING
9	295A9601 P3	EXHAUST FRAME

Figure MI-D1. Guide Pin Locations

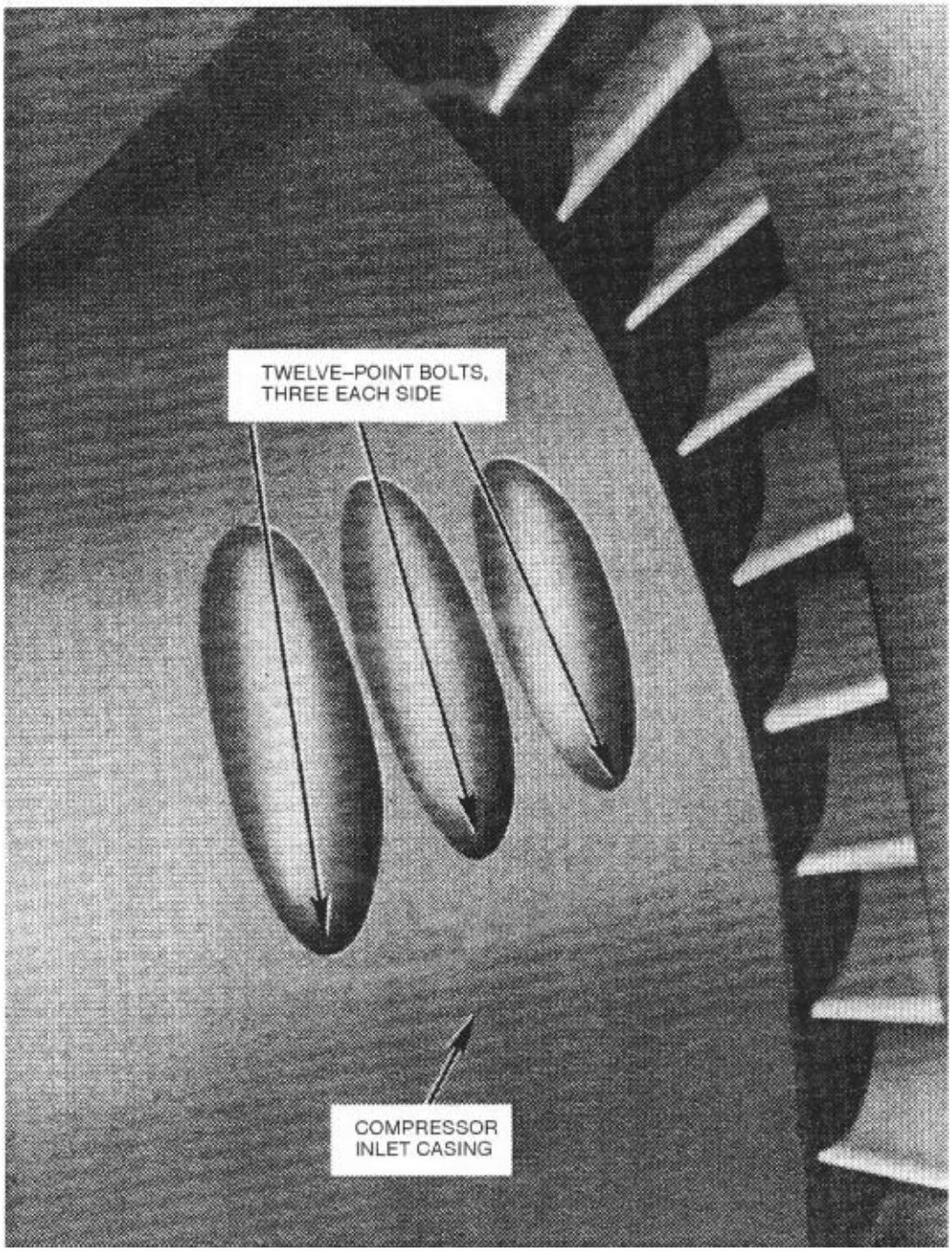


Figure MI-D2. Twelve-Point Bolts in Inlet Casing

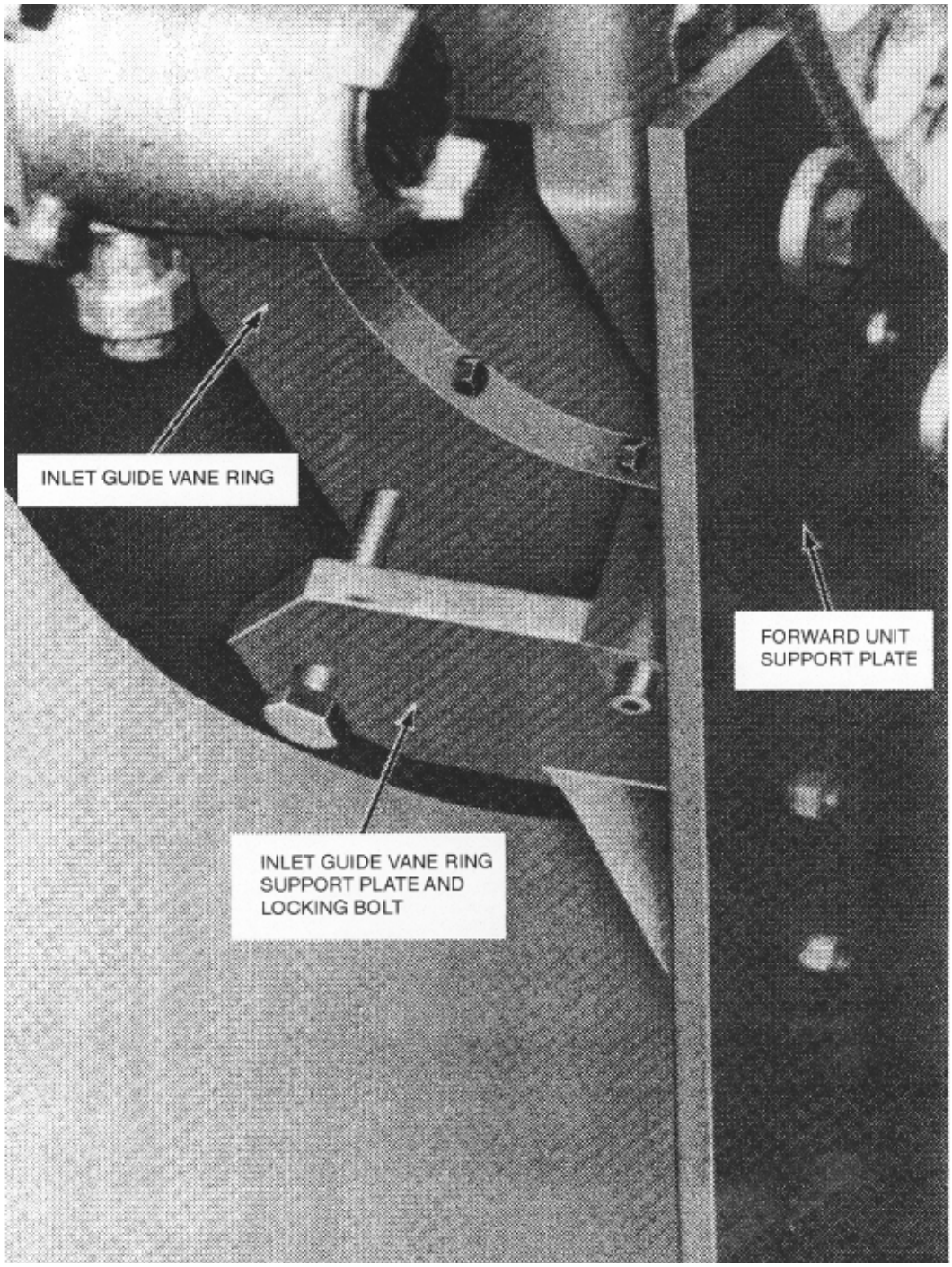


Figure MI-D3. IGV Ring Support Plate and Locking Bolt

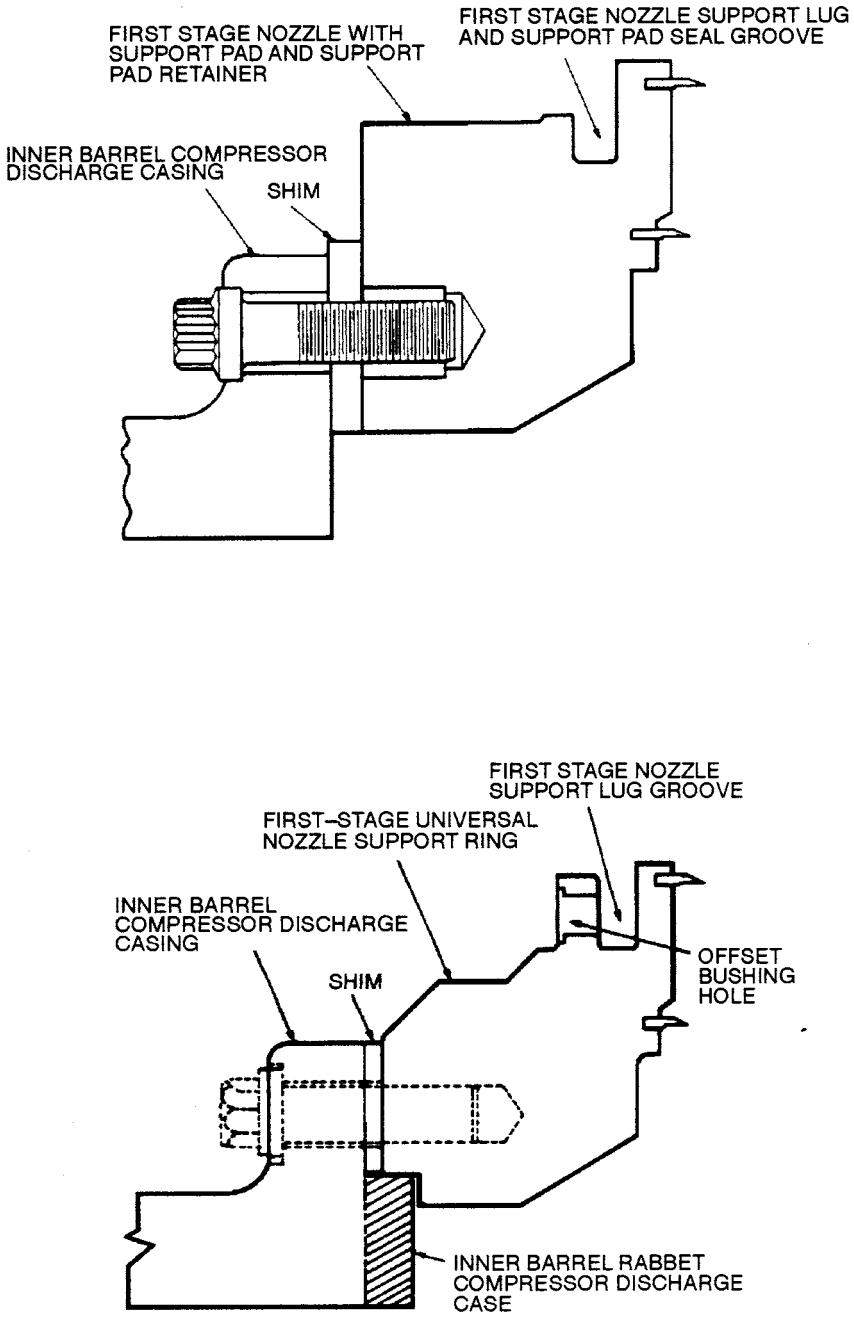


Figure MI-D4. First Stage Nozzle Support Ring and Support Groove

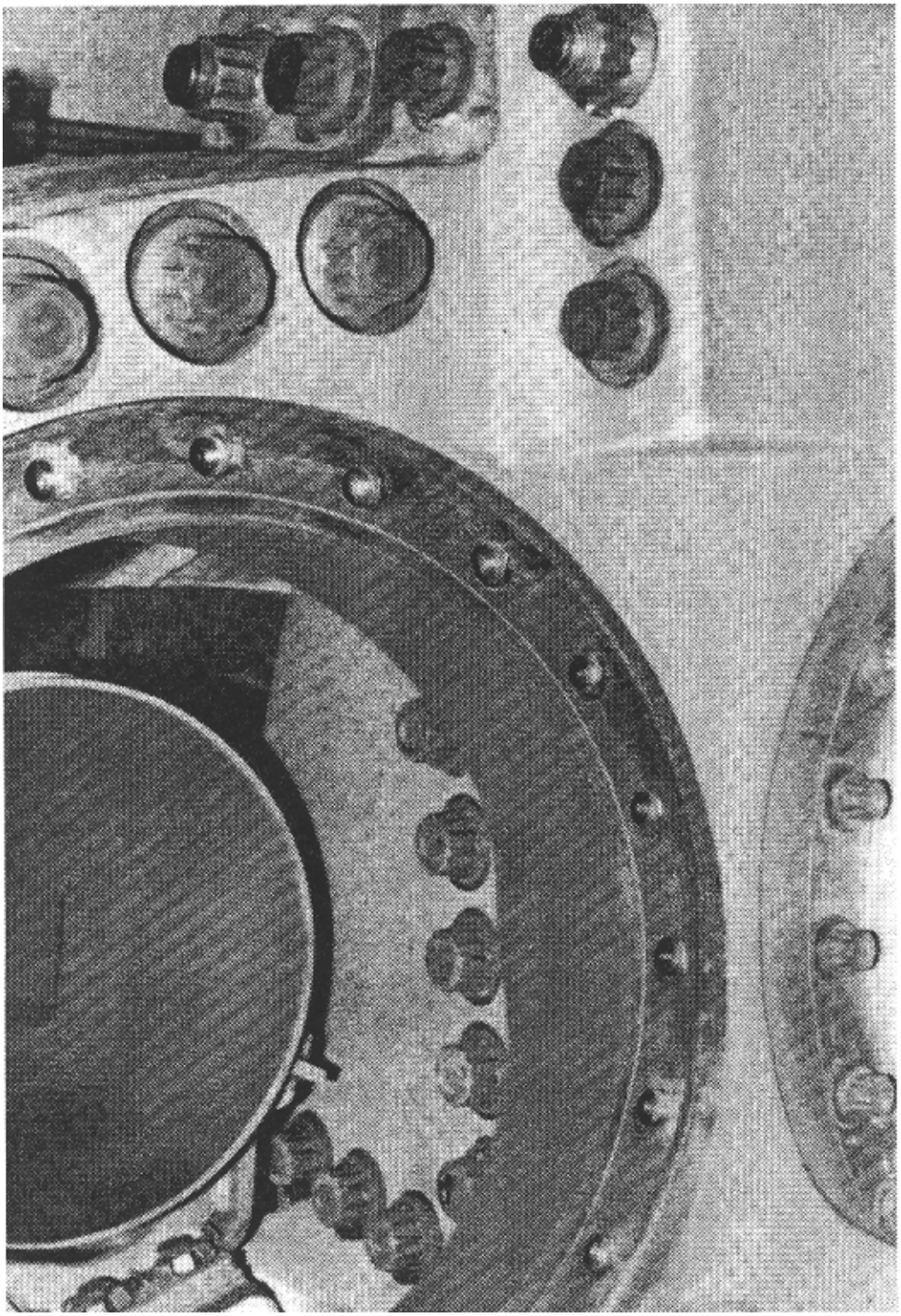


Figure MI-D5. Inner and Outer CDC Bolts

OPERATION 13 - REMOVE COMPRESSOR DISCHARGE AND AFT COMPRESSOR CASING

NOTE Later vintage units do not have an aft compressor casing, it has been combined as a one piece casing with the forward compressor casing. If the unit does have an aft compressor casing it can be removed along with the compressor discharge casing as a single unit.

1. Remove the vertical flange bolting between the upper halves of the compressor discharge casing and inner compressor discharge casing.

2. Remove all the horizontal joint bolting from the compressor discharge casing inner and outer joints. See Fig. MI-D5.

3. Remove the horizontal bolting from the aft compressor casing joint (if applicable).

4. Remove the circumferential bolting between the aft flange of the compressor casing and the forward flange of the first stage nozzle support ring.

5. Insert the guide pins, eight total, in the body bound bolt holes of the aft compressor casing, if applicable, and the compressor discharge casing. Refer to Special Tool Listing of this Service Manual for proper guide pins to be used. See Fig. MI-D1 for guide pin locations in the casings.

6. Jack the aft, if applicable, and compressor discharge casings apart utilizing the jacking bolt holes provided. Refer to Special Tool listing in this Service Manual for proper jacking bolts to be used in each casing.

10. Rig and lift the compressor discharge and aft compressor casing, if applicable, clear of the unit. Refer to the Weights and Center of Gravity drawing in the Service Manual.

11. Set the compressor discharge casing on suitable blocking in an upright position on the inner vertical bolting flange with supports placed between the compressor discharge bulkhead vertical flange and the seating surface. See Fig. MI-D6.

OPERATION 14 - REMOVE THE INNER COMPRESSOR DISCHARGE CASING

1. Unbolt the recessed horizontal joint bolts in the inner compressor discharge casing. Install the lifting lugs and lift the top half clear. Exercise care when starting to lift as there are no jack bolts to separate the joint. See Figures MI-D7 and MI-D8.

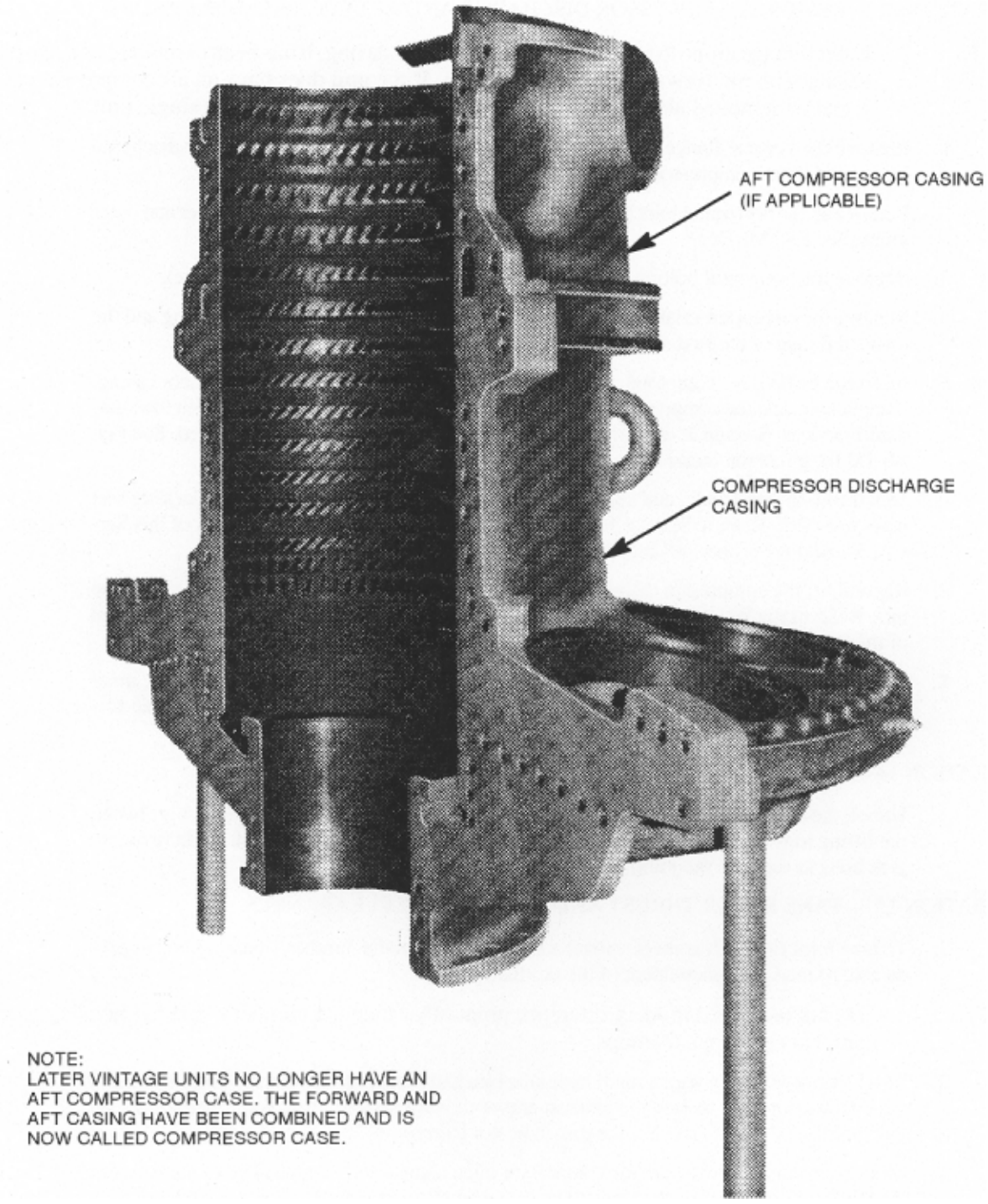


Figure MI-D6. CDC and AFT Compressor Casing Supported in Upright Position

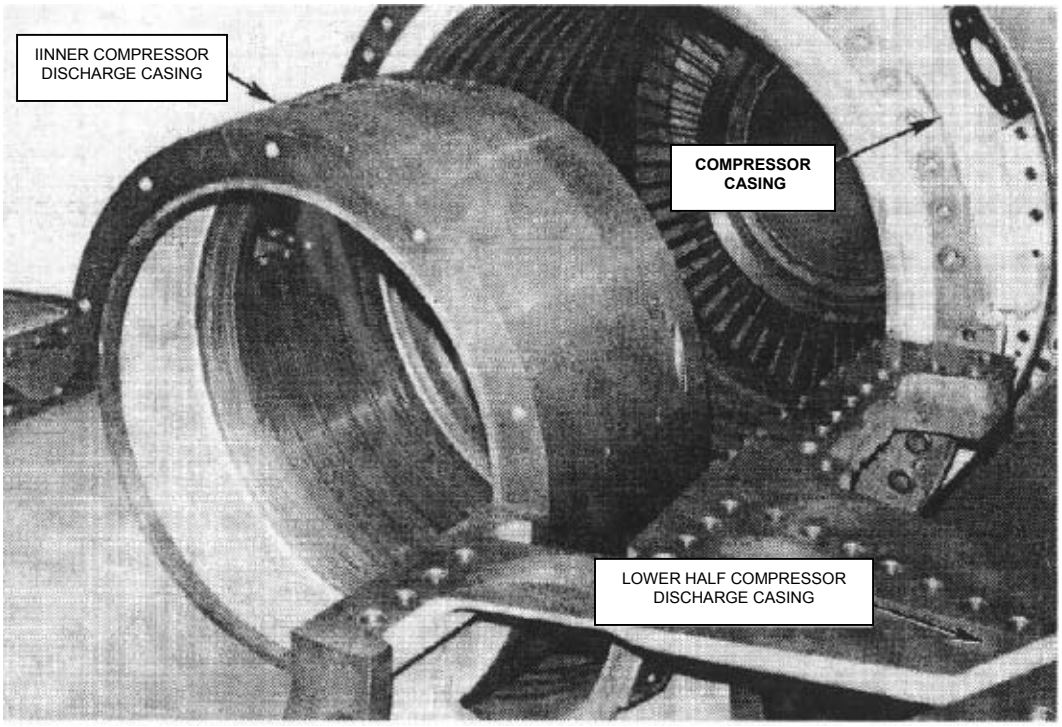


Figure MI-D7. Inner CDC at Initial Assembly

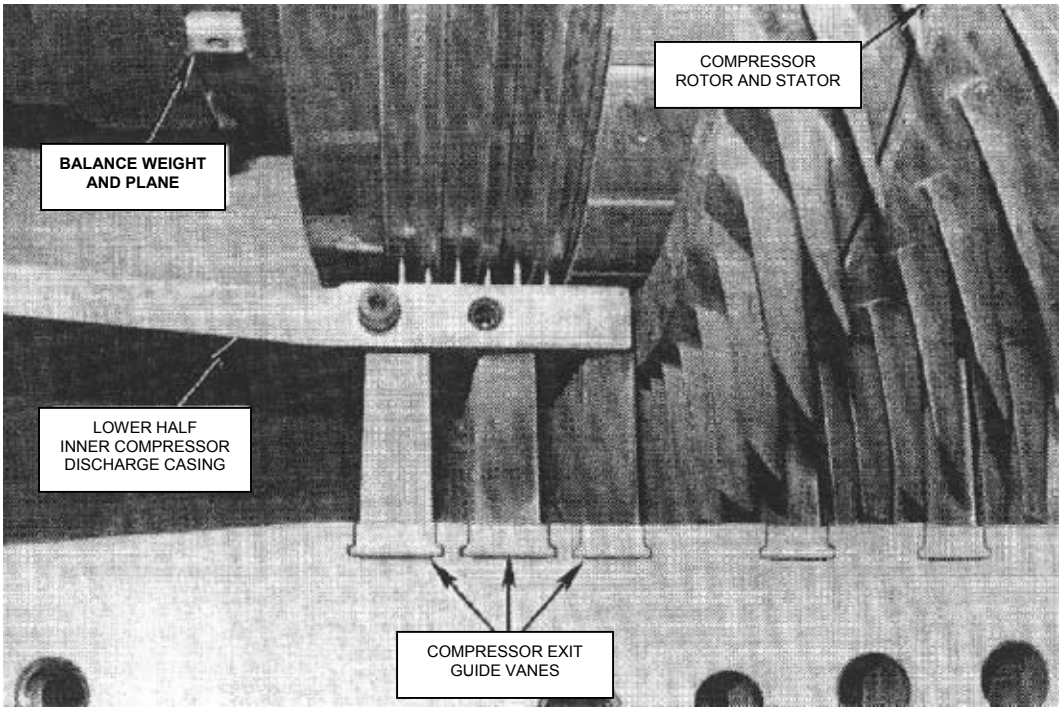


Figure MI-D8. Inner CDC with Upper Half Removed

OPERATION 15 - TAKE ROTOR THRUST AND COMPRESSOR CLEARANCES

1. To take rotor thrust clearances, mount a dial indicator on the number 1 bearing housing, to be able to read axial movement of the turbine rotor.

   NOTE Do not use any stationary reference point other than the number 1 bearing housing so as to prevent erroneous readings.

2. Move the rotor axially with a small hydraulic ram and blocking between the inlet casing and the compressor rotor toward the exhaust end of the turbine until the rotor "bumps" against the "unloaded" thrust face. Set the dial indicator to zero.

3. Move the rotor axially toward the compressor inlet, using a small ram and blocking between the load gear coupling and the forward load gear bearing housing. The rotor should be firmly against the "loaded" thrust face but not loaded so as to deflect the bearing or casings when the indicator is read.

4. Read the amount of rotor movement as shown on the dial indicator.

5. The range of axial movement of the rotor without distorting the rotor or bearing mounts is usually between 0.014 and 0.020 inch. Repeat the rotor thrust check at least three (3) consecutive times in order to assure repeatability.

6. Record thrust movement and compressor clearances on Inspection Form GT7200.

   NOTE All axial compressor clearances are to be measured with the rotor in contact with the forward or "normally loaded" face of the thrust bearing.

7. Measure and record stator blade tip clearances for all seventeen compressor stages (S1 to S17).

8. Measure and record compressor rotor blade tip clearances for all seventeen compressor stages (R1 to R17).

9. Measure and record exit guide vane stator blade clearances (EGV1 and EGV2).

10. Measure and record the compressor rotor leak off clearances as well as the radial and axial packing gland clearances (P1, P2, P3 and P4).

11. Measure and record the inlet guide vane and rub ring clearances (X, X1, and X2). Clearances are recorded on Inspection Form GT7200. Refer to MLI 0404 for clearance specifications.

NOTE If dirt has been left on compressor blade tips after cleaning during operation and prior to shutdown it may be necessary to hand clean the blade tips before making clearance checks or proceeding with visual inspection. Refer to Gas Turbine Compressor Cleaning in the Service Manual.

12. Visually inspect compressor for blade tip rubs, cracks, corrosion, and foreign object damage. Report conditions found on Inspection Form GT4001 and GT4055.

OPERATION 16 - PERFORM HOT GAS PATH DISASSEMBLY PROCEDURES 12 THROUGH 15

NOTE Refer to Section 4B, Hot Gas Path Inspection, in this Service Manual for the following disassembly operations.

1. Remove lower half first stage nozzle eccentric pin and horizontal nozzle clamps.

2. Remove lower half first stage nozzle.

3. Remove lower half second and third stage nozzle radial retaining pins.

4. Remove lower half second and third stage nozzle segments.

OPERATION 17 - REMOVE UPPER HALF NUMBER ONE AND TWO BEARING HOUSINGS

NOTE The upper half of number one bearing housing weighs approximately 560 pounds and the upper half of number two bearing weighs approximately 380 pounds.

Number One Bearing

NOTE On some units the upper half number one bearing liner will be assembled to the upper half housing and will be removed with the housing. Later vintage units the upper half bearing liner is bolted to the lower half bearing liner.

1. Disconnect and tag any instrumentation on the upper half bearing housing.

2. Loosen and remove horizontal joint bolting on the number one bearing housing. See Fig. MI-D9.

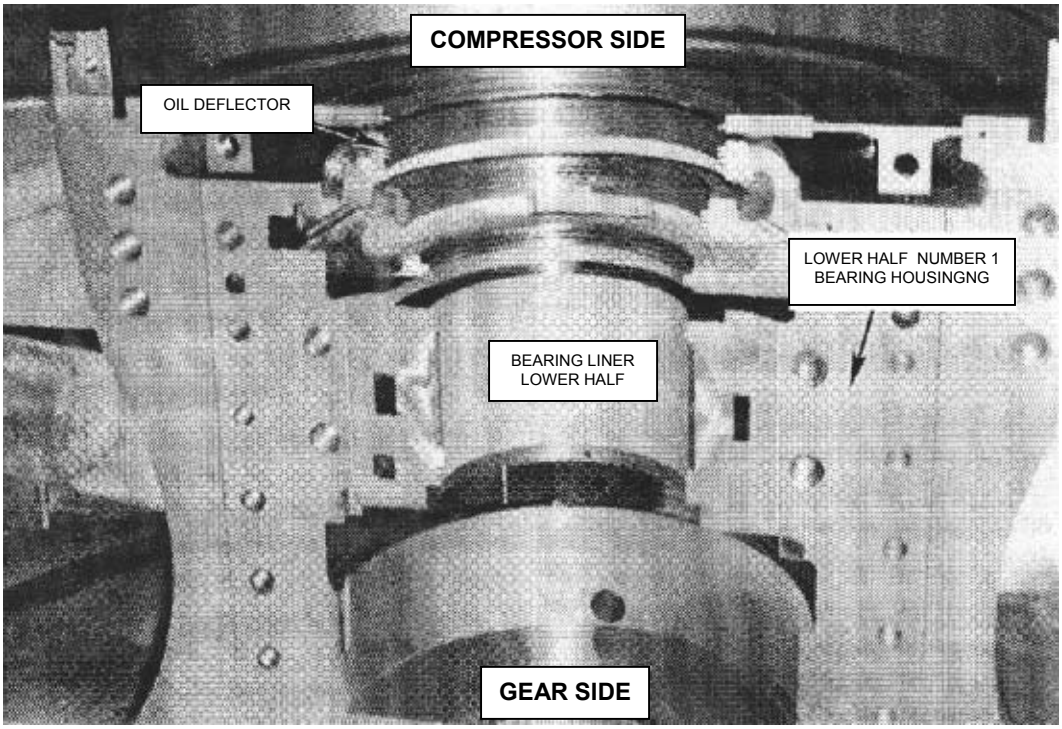


Figure MI-D9. No. 1 Bearing Housing with Upper Half Removed

3. Install jacking bolts and jack the housing apart 0.100 inch or less to separate the horizontal joint.

4. Rig to lift the housing, using a chainfall, and lift the housing clear of the unit. Set the housing on suitable blocking or plywood to protect from damage.

5. Unbolt and remove the upper half bearing liner from the lower half liner (if applicable).

6. Cover the open bearing housing to protect the rotor bearing journal and to keep the housing clean.

7. Identify wrap and/or cover the removed parts for protection.

Number One Bearing Housing Thrust Bearings

1. Remove the active tilted pad thrust bearing from the bearing housing by carefully sliding out the top three pads, rotating the cage to the split line, and lifting off the top of the cage.

2. Roll out the lower half cage and remove each pad as they become accessible.

3. Identify and tag the parts removed for reassembly.

4. Wrap the thrust bearing with protective wrap to prevent damage to parts.

5. Remove the upper and lower half tapered land inactive thrust bearing.

6. Identify and tag the parts removed for reassembly.

8. Wrap the thrust bearing with protective wrap to prevent damage to parts.

9. Cover the open lower half bearing housing to protect the bearing journal and keep the housing clean.

Number Two Bearing

NOTE The number two bearing is accessible for disassembly and reassembly during major inspection since the exhaust frame is disassembled, however special fixturing (tooling) is provided to disassemble and reassemble the number two bearing with the exhaust frame in place.

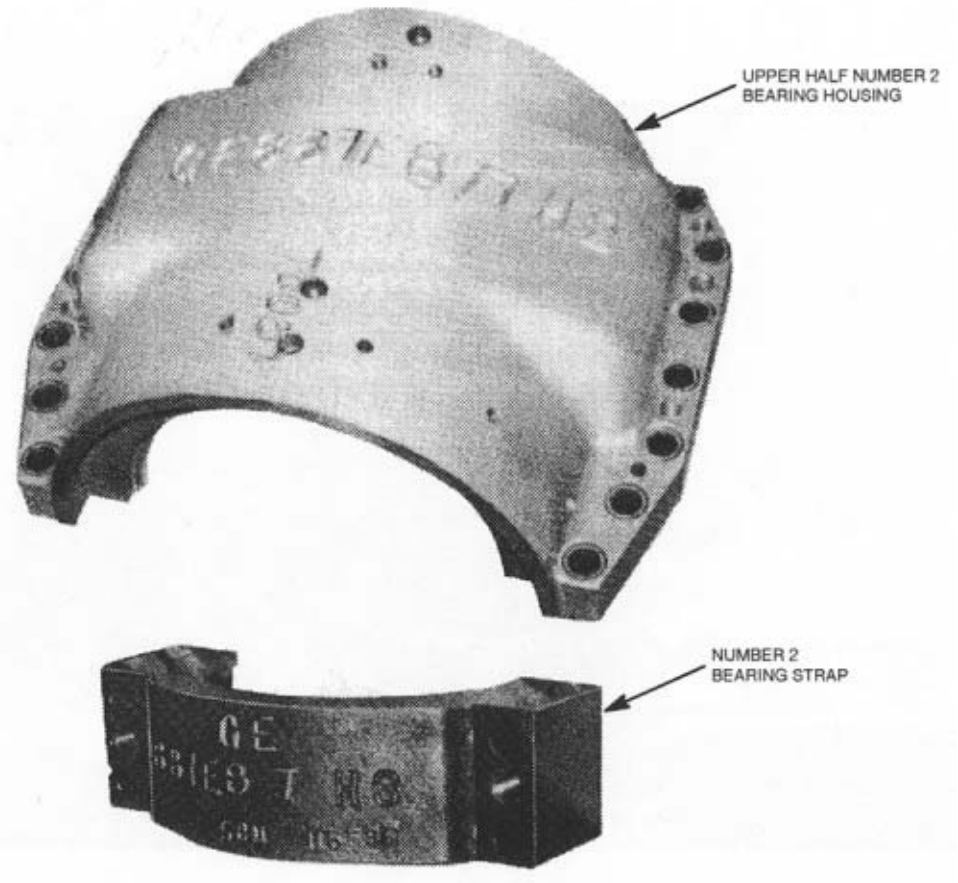


Figure MI-D10. No. 2 Bearing Strap and Upper Half No. 2 Bearing Housing

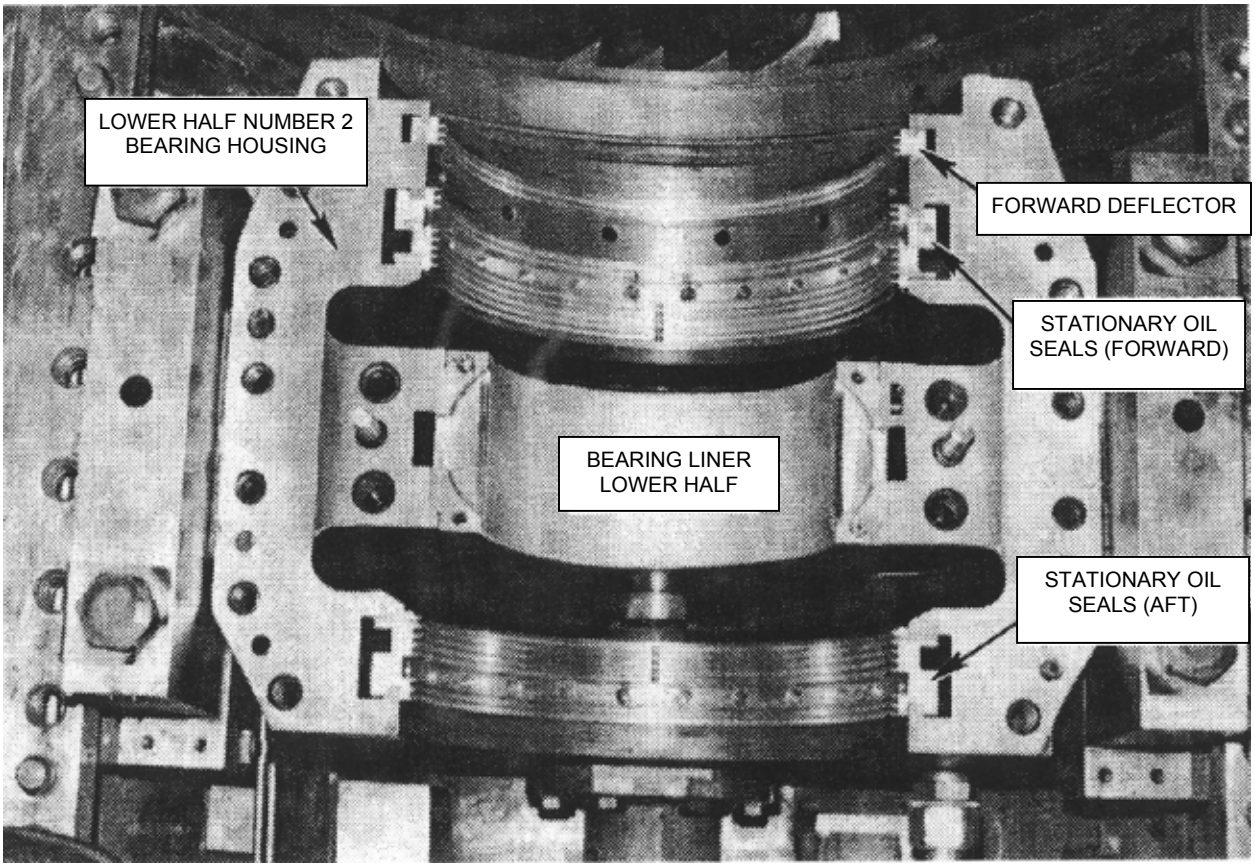


Figure MI-D11. No. 2 Bearing Housing with Upper Half Removed

1. Loosen and remove the horizontal bolts from the bearing housing. See Figs. MI-D10 and MI-D11.
2. Using jack bolts, jack the housing apart to separate the horizontal joint.
3. Rig to lift the housing, using a chain fall, and lift the housing clear of the unit.
4. Unbolt and remove the top half of the number two bearing exhaust baffle.
5. Unbolt and remove the bearing strap and upper half bearing liner.
6. Cover the open bearing housing to protect the rotor bearing journal.
7. Wrap the removed liner for protection, identify and tag all parts for reassembly.

OPERATION 18 - PERFORM HOT GAS PATH DISASSEMBLY OPERATIONS 16 AND 17

NOTE Refer to EM 5321, Hot Gas Path Inspection, in this Service Manual for the following disassembly operations.

1. Remove upper half second and third stage nozzle radial retaining pins.
2. Remove upper half second and third stage nozzle segments.

INSPECTION

OPERATION 1 - INSPECT COMBUSTION SYSTEM COMPONENTS PER COMBUSTION INSPECTION OPERATIONS 1 THROUGH 7 AND OPERATIONS 9 AND 10

NOTE Refer to EM 5320, Combustion Inspection, for the following disassembly operations.

1. Pressure test fuel oil check valves.
2. Inspect and test fuel nozzles.
3. Inspect combustion liners.
4. Inspect crossfire tubes and spring retainers
5. Inspect transition pieces.
6. Inspect forward and aft flow sleeves.
7. Inspect combustion outer casings.
8. Inspect and test spark plugs.
9. Inspect and test flame detectors.

OPERATION 2 - INSPECT HOT GAS PATH COMPONENTS PER HOT GAS PATH INSPECTION OPERATIONS 2 THROUGH 6

NOTE Refer to EM 5321, Hot Gas Path Inspection, for the following disassembly operations.

1. Inspect first, second and third stage nozzles.
2. Inspect first, second and third stage buckets.
3. Inspect shroud blocks.
4. Inspect second and third stage nozzle diaphragms.
5. Make first stage nozzle ellipticity checks.

OPERATION 3 - INSPECT INLET GUIDE VANES

1. Inspect the inlet guide vanes carefully for deposits, erosion (thinning of trailing edges and rounding of leading and trailing tip corners) or corrosion pitting.

2. Liquid penetrant inspect for cracks in the blade root and vane.

3. Two samples of blade deposit, if any is found, should be collected in a clean glass jar or bottle and sent to a testing laboratory for analysis. This analysis is important to the overall inspection.

4. Check for oily substances on the inlet guide vanes.

5. Record inspection results on Inspection Form GT4055.

OPERATION 4 - INSPECT COMPRESSOR ROTOR, STATOR BLADING AND COMPRESSOR AND TURBINE CASINGS

1. Inspect blading for tip bending damage. Some bending damage may be field repaired under supervision of your GE Field Service Representative.

2. Inspect the air foil surfaces for nicks and dents. Small, shallow dents above the pitch line may be blended out. Damage below the pitch line is not acceptable and blades must be replaced.

3. Perform liquid penetrant check after working on blading.

4. Check blading for erosion, corrosion, and deposits. Samples of deposits should be collected in glass containers for analysis. Blading eroded beyond about 20% of root area must be replaced.

5. Liquid penetrant check blade roots and airfoils. Damage to blading which requires replacement should be performed by an authorized GE Repair Facility.

6. Inspect all casings for cracks or other physical damage.

7. Record inspection results on Inspection Form GT4001.



If inspection of the compressor or turbine indicates a need to remove the compressor and turbine rotor to inspect the lower half components it is recommended you contact your GE Field Service Representative.

OPERATION 5 - INSPECT FIRST STAGE NOZZLE SUPPORT RING

1. Check the nozzle support ring for rotor seal damage or wear.

2. Inspect the nozzle support ring groove for galling or fretting.

3. Report inspection findings on Inspection Form GT9350.

OPERATION 6 - INSPECT INNER COMPRESSOR DISCHARGE CASE SEALS

1. Inspect the seals for damage, wear and overall integrity.

2. Report any discrepancies to your GE Field Service Representative.

OPERATION 7 - INSPECT SLEEVE BEARINGS AND THRUST BEARINGS

1. Assemble rotor support and lever in exhaust frame and under the shaft to raise the shaft for the removal of the lower half number two bearing liner. See Fig. MI-I1.

2. Mount a dial indicator at the bearing liner centerline, using the rotor support lever, raise the shaft 0.014 inch maximum and block into position with rotor support lever threaded bolt.



Take appropriate precautions to assure that the shaft is not damaged by the rotor support lever when raising the shaft.

3. Roll out the lower half number two bearing liner.

4. Place protective covering over the bearing housing to protect the rotor bearing journal and keep foreign matter out of the housing.

5. Inspect journal bearing babbitt surface for cracks, wiped babbitt, scoring, foreign material, pitting, spalling, and excessive or abnormal wear patterns. Cracked and chipped babbitt can generally be repaired in an approved GE Repair Facility, if the damage is not excessive. The cause of the damage within the unit must be eliminated to prevent further cracking and chipping.

NOTE Lube oil contamination with dirt or metal particles will result in scored surfaces of rotor journals, thrust faces, bearing liners, and thrust bearings. A characteristic of particle scored bearings is the frequent occurrence of full circumferential score marks on the rotating members. On journal and thrust bearing liners the score marks will frequently extend through the circumferential length of the pressure zone. Although damage to bearing surfaces will result in a theoretical performance degradation, certain practical limits can be set within which the function of the bearing is not compromised beyond the degree normally expected from manufacturing variables. A restricted number of circumferential grooves can therefore be tolerated in the bearing area provided the total volume of the groove (which acts as a partial oil short-circuit) is limited. The following are acceptance limits which can be used as an acceptance guide for used bearing components.

6. Some scratching or pitting will always occur on babbitted bearing surfaces during normal operation because of impurities which accumulate in the lubricating oil system. Scratches or pits that are not over 0.005 inch deep, pits that are not over 0.100 inch in diameter and pits or scratches do not extend axially more than 1/3 of the bearing length, (axial scratches and pit formations must be circumferentially at least 45° apart), are acceptable. Raised metal is not allowed.

   NOTE The cumulative width of all scratches in circumferential and axial directions should not exceed 0.100 inch. (Circumferential or axial pit formations exceeding three in number must be counted as continuous scratches.) Raised metal may be removed by scraping or lightly polishing the affected area with plastic wool such as "Scotchbrite." Damage that exceeds the specified limits requires replacement of the bearing liner.

7. If the bearing liner has experienced excessive temperatures, the babbitt will show evidence of smear and wiping. If the wiped area is less than five percent of the liner lower-half area, the bearing may be reused after re pair of the wiped surface. Larger wiped areas must be repaired by an approved GE Repair Facility. Before replacing the liner, the cause of the wipe must be determined and corrected.

8. Most journal bearings develop polished areas during low speed cranking or coastdown conditions. The polished area is usually found at the bottom centerline of the bearing liner and should extend along the entire length of the bearing. This type of pattern is normal and should not be cause for any alarm. If, however, the polished area is found only on one edge of the liner, or if it appears on the liner upper half, a problem is indicated which must be corrected to prevent eventual bearing wipe.

9. Measure journal diameter at two planes, fore and aft. Record these dimensions and the general condition of the journals on Inspection Forms GT2115 and GT7200. Roundness and taper are two of the most critical dimensions associated with a bearing journal. These dimensions are established with a four point check taken in the vertical and horizontal planes (at 90° to one another) at both the forward and aft edge of the journal. If the journal diameter is 0.002 inch or more outside of its drawing tolerance, and, if the liner-to-journal clearance falls outside of the drawing tolerances, it may be necessary to remachine the journal. It should be noted that whenever a journal is remachined, the rotor must be rebalanced. Journal tapers that fall outside of the specified drawing tolerances may also require remachining and rebalancing. Scratches or pits that are not over 0.005 inch deep, pits that are not over 0.100 inch in diameter and pits or scratches that do not extend axially more than 1/3 of the bearing width, (axial scratches and pit formations must be circumerentially at least 45° apart) are acceptable. Raised metal is not allowed.



The cumulative width of all scratches in circumferential and axial direction should not exceed 0.100 inch. (Circumferential or axial in-line pit formations exceeding three in number must be counted as continuous scratches.) Raised metal may be removed by lapping, stoning, or lightly strapping the entire journal surface. Damage that exceeds the specified limits may be repaired by an approved GE repair facility by skim cutting the journal to a diameter 0.002 inch below drawing minimum. Deeper imperfections must be repaired by turning the journal to the next appropriate undersize diameter specified by an approved GE repair facility.

10. Clamp the number one (1) and the number two (2) bearing halves together, and make a six point check for inside diameter. Record all bearing and journal measurements and condition on GT7200. Record all bearing seal clearances and condition on Inspection Forms GT7200.

11. Coat the lower half number two bearing liner to be installed with clean turbine oil used in the unit lube system and set the liner on the rotor journal. Roll the bearing liner carefully into the lower half of the bearing housing. Restrain liner from falling into the lower half.



To avoid deleterious effects, use of additives with gas turbine lubricant or in areas wetted by the lubricant should be limited to those approved by the fluid supplier or by GE Engineering. If there are questions or doubts about a material, contact your GE Field Service Representative.

12. With the lower half number two bearing liner installed and positioned, carefully release the rotor support lever and gently lower the rotor journal to rest on the lower bearing liner.

13. Recover the number two bearing housing to protect the bearing journal and keep foreign matter out of the housing.

14. Support the rotor in the number one bearing area with a suitable jack to raise the shaft for the removal of the lower half number one bearing liner.

15. Mount a dial indicator at the bearing liner centerline, using the jack raise the shaft 0.014 inch maximum to release the pressure on the bearing liner.

16. Repeat disassembly and inspection Steps 3 through 10 for the number one bearing liner.

17. Coat the lower half number one bearing liner to be installed with clean turbine oil used in the unit lube system and set the liner on the journal. Roll the bearing liner carefully into the lower half of the bearing housing. Restrain the liner from falling into the lower half.

18. With the lower half number one bearing liner installed and positioned, carefully release the rotor support jack and gently lower the rotor journal to rest on the lower bearing liner.

19. Recover the number one bearing housing to protect the bearing journal and keep foreign matter out of the housing.

20. Thrust bearings just as with journal bearings, the babbitt surfaces must be inspected for cracks, chips, scratches,and wiping. Cracks or chips must be eliminated before the turbine is placed back in operation. Scratches or pits that are not over 0.005 inch deep, pits that are not over 0.100 inch in diameter and pits or scratches that do not extend radially more than 1/2 inch, one per pad are acceptable. Raised metal is not allowed.



The cumulative width of all scratches in circumferential and radial directions should not exceed 0.100 inch. (Radial or axial inline pit formations exceeding three in number must be counted as continuous scratches.) Raised metal may be removed by scraping or lightly polishing the affected area of the pad surface with plastic wool such as "Scotchbrite". Damage that exceeds the specified limits requires replacement of the affected pad(s) if in a self adjusting tilting-pad thrust bearing. The entire bearing must be replaced in the case of a fixed tilting-pad thrust bearing or fixed arc thrust bearing. Babbitt wipes that occur in an area representing ten percent of less of the respective pad or land area can be repaired. Wiping that extends over larger areas, requires re-babbitting of the pad or land. It will be necessary to determine the cause of the wiping prior to reinstalling the bearing. Any repairs should be done by an approved GE repair facility.

21. Many new gas turbines are employing tilting pad thrust bearings with oil dam inserts. These oil dams serve to properly direct the oil onto, and away from, the respective bearing pads. The axial clearance between the tops of the oil dams and the babbitt surfaces is a

critical dimension and must be maintained within the drawing tolerances to assure proper bearing operation.

22. Satisfactory operation of tapered land bearings is very dependent upon the amount of taper present in the bearing face. If the measured taper falls outside of drawing tolerances, corrective action must be taken before continued use of the parts.

23. Inspect thrust runners for galling, pitting, scoring and excessive wear. It is necessary that the thrust runner face be perpendicular to the rotor bearing journals. Scratches or pits that are not over 0.005 inch deep, pits that are not over 0.100 inch in diameter and pits or scratches that do not extend radially more than 1/2 inch in length, (radial scratches or pit formation must be circumferentially at least 60° apart) are acceptable. Raised metal is not allowed. Record findings on Inspection Forms GT2150 and GT7200.



The cumulative width of all scratches in circumferential and radial directions should not exceed 0.100 inch. (Radial or circumferential inline pit formations exceeding three in number must be counted as continuous scratches.) Raised metal may be removed by lapping, stoning, or lightly polishing the entire thrust runner surface with carborundum paper backed up with a flat block. Damage that exceeds the specified limits may be repaired by machining the thrust runner face to below defect depth but not below 90 percent of the drawing minimum thrust runner thickness.



It is recommended that remachining operations be accomplished under the guidance of a qualified GE Field Service Representative or at an approved GE repair facility.

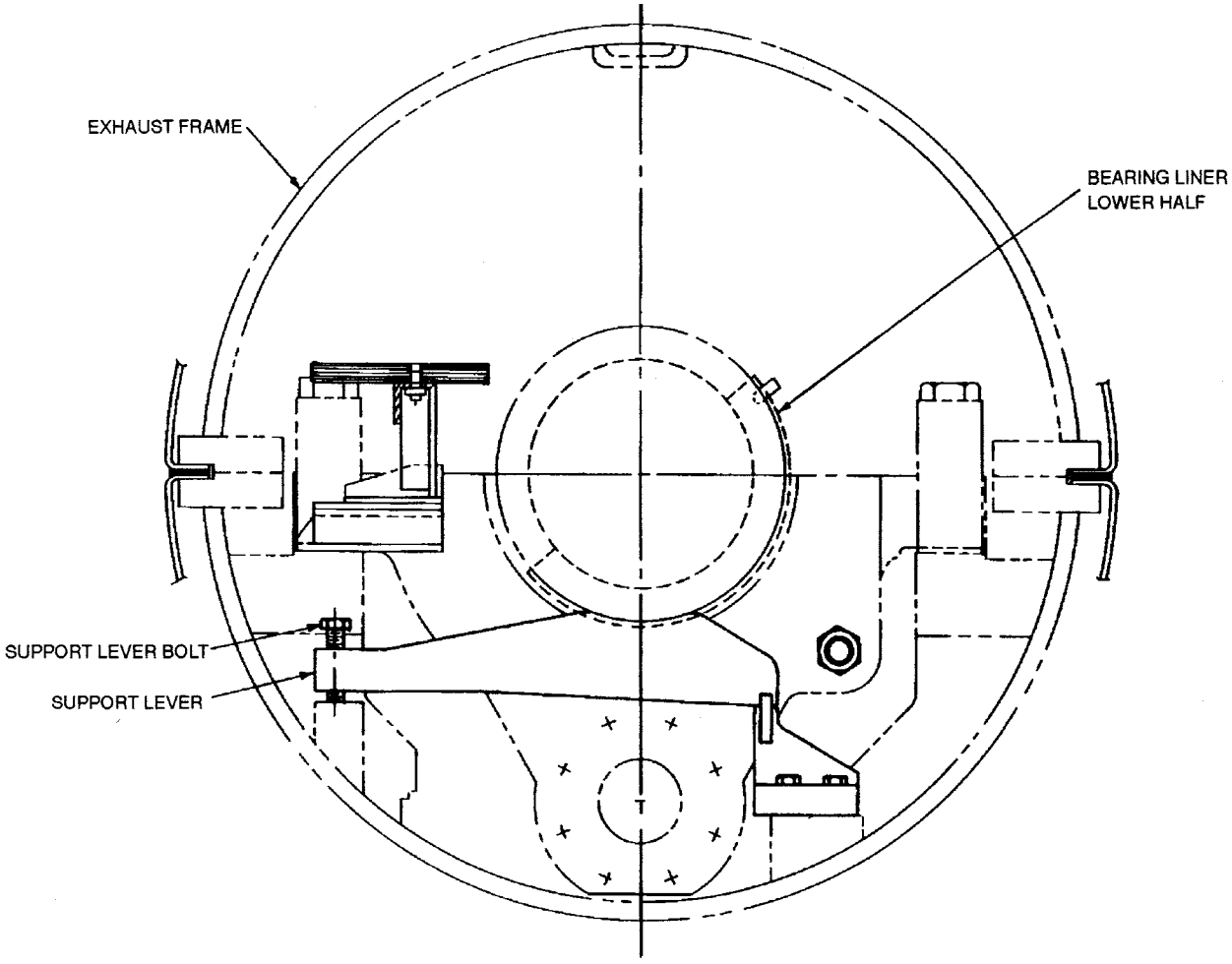


Figure MI-I1. Rotor Support Lever for Removal of No. 2 Bearing Liner Lower Half

REASSEMBLY

OPERATION 1 - REASSEMBLE SLEEVE BEARINGS, THRUST BEARINGS, UPPER HALF BEARING HOUSINGS AND EXHAUST HOOD BAFFLE

Caution **Prior to reassembly of thrust bearings and bearings, carefully check the housings and bearing areas to assure all tools, parts and debris have been removed and the housings are thoroughly clean before closing.**

NOTE Use a very light coating of clean, turbine oil only during assembly of the thrust bearings and bearings.

1. Reassemble the thrust bearings in the reverse manner as they were disassembled. Roll in the inactive thrust lower and upper halves and position the anti-rotation pin.

2. Assemble the inactive thrust shims into the bearing cavity.

3. With inactive thrust bearing and shims assembled in place thrust the rotor up against the inactive thrust bearing to allow room to assemble the active thrust bearing and shims.

4. Roll in the lower half of the active thrust bearing base ring, installing the thrust pads as it enters the lower housing. Set the upper half of the base ring on the active thrust bearing and install the upper half thrust pads. Roll the assembly around to position the anti-rotation key. After the active thrust bearing is assembled, slide the rotor position shim into place behind the active thrust bearing.

5. Check the rotor journal and number one bearing housing lower half for cleanliness.

6. Assemble the upper half bearing liner to the upper half bearing housing or to the lower half bearing liner, whichever is applicable.

7. Clean and deburr the mating joints of the number one bearing housing. Apply a light coating of joint sealant to the lower horizontal joint. Refer to Section 2, Standard Practices, recommended solvents, sealers and cleaners in this Service Manual for proper joint sealants.

8. Rig, using a chainfall, and lift the upper half housing into position on the lower half. Apply a coating of anti-seize compound to dowels and bolting. Install bolting and torque bolts alternately from the center of each horizontal joint outward.

9. Reconnect all instrumentation and tubing which was removed.

10. Assemble the upper half exhaust air cone (baffle) to the lower half in the number two bearing area.

11. Check the rotor journal and the number two bearing housing for cleanliness, clean and deburr bearing strap mating surfaces and number two bearing housing horizontal joints.

12. Check the anti-rotation pin and install the upper half number two bearing liner.

13. Apply joint sealant to bearing strap and install bearing strap shims and bearing strap.

14. Lightly coat the dowel pins and bolts with anti-seize compound, install dowel pins, bolts and torque alternately.

15. Apply joint sealant to number two bearing housing, install upper half bearing housing to the lower bearing housing.

16. Lightly coat dowel pins and bolts with anti-seize compound, install dowel pins and bolts. Torque bolts alternately from the center of each flange joint outward.

17. Reconnect any instrumentation and tubing which was removed.

OPERATION 2 - RECHECK COMPRESSOR CLEARANCES AND THRUST



The thrust bearings must be fully assembled when taking thrust measurements. Any movement of the unit rotor without fully assembled thrust bearings will cause the unit journals to move axially in the bearing liners causing the compressor blades to engage the stator blades. Such contact can cause major blade damage.

1. Recheck all thrust and compressor clearances. Refer to disassembly operation 15, Steps 1 through 11 for clearances to be taken.

2. Record the thrust and clearance measurements on Inspection Form GT7200

OPERATION 3 - INSTALL INNER COMPRESSOR DISCHARGE CASING



Inspect the inner case for cleanliness and foreign matter prior to assembly of the top half casing.

1. Clean and deburr the horizontal flange on the lower half casing.

2. Ensure the four dowel pins (two forward and two aft) are assembled in the horizontal joint and the tapped bolt holes are clean.

3. Rig to lift the top half inner compressor discharge casing. Lift the casing, clean and deburr the horizontal joint flange.

4. Set the upper half inner compressor casing in place and assure that all dowels are in place.

5. Apply a thin film of anti-seize compound to the bolts, install bolting and torque the horizontal joint.

6. Check the outside diameter of the upper and lower half casings at the horizontal joint to ensure that the mating joints do not have a step greater than 0.0015 inch.

7. Using a piece of 0.0015 inch feeler check the aft end inner bore horizontal joint at the rabbet fit. This joint fit should not exceed 0.0015 inch.

OPERATION 4 - INSTALL COMPRESSOR DISCHARGE AND AFT COMPRESSOR CASINGS

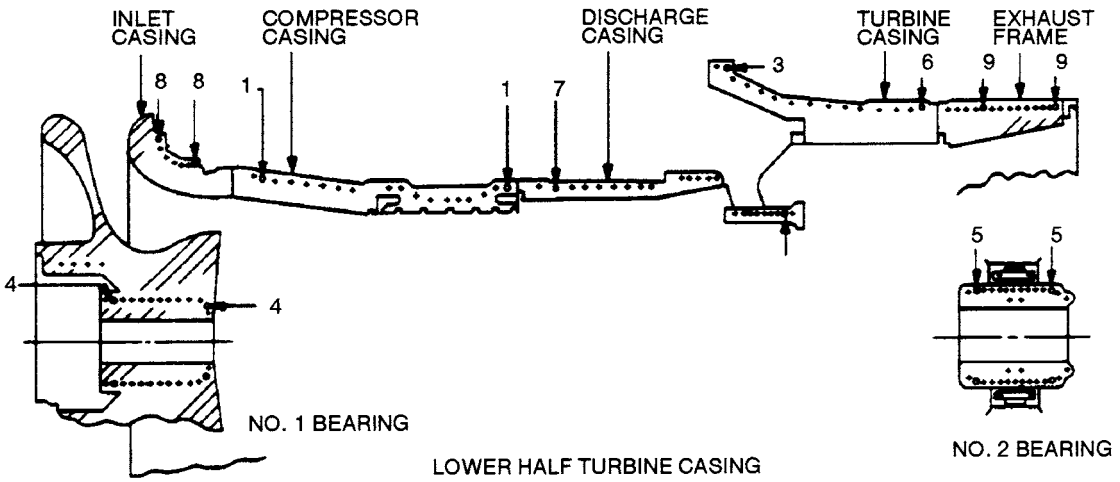


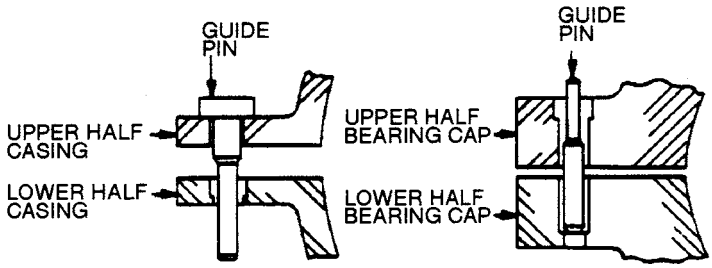
Later vintage units do not have an aft compressor casing, it has been combined as a one piece casing with the forward compressor casing. If the unit does have an aft compressor casing it can be installed along with the compressor discharge casing as a single unit.

1. Clean and deburr all mating joints.

2. Rig to lift the compressor discharge and aft compressor casing. Install the guide pins. See Figure MI-R1.

3. Apply joint compound on the surface of the horizontal joints only and only for a distance of one inch on both sides of the four way joint. Do not apply joint compound to the vertical joint. Refer to Section 2, solvents, sealers and cleaners in the Service Manual.





NOTES: GUIDE PINS FOUND IN TOOL KIT, MAJOR DISASSEMBLY

GUIDE PINS ARE USED IN BODY ROUND STUD HOLES OF CASINGS & DOWEL HOLES OF BEARING CAPS.

ITEM	GUIDE PIN	WHERE USED
1	248A4880 P4	COMPRESSOR CASING
3	295A9609 P2	TURBINE CASING
4	295A9601 P1	NO. 1 BEARING CAP
5	295A9601 P2	NO. 2 BEARING CAP
6	295A9609 P3	TURBINE CASING
7	295A9609 P1	DISCHARGE CASING
8	295A9600 P1	INLET CASING
9	295A9601 P3	EXHAUST FRAME

Figure MI-R1. Guide Pin Locations

4. Carefully lower the casing into position on the lower half casings using the guide pins installed.

5. Apply anti-seize compound to all bolting.

6. Remove guide pins and install bolting to the horizontal joints.

7. Tighten the horizontal joint bolts starting at the mid point of the horizontal joints. Torque the bolting working alternately left and right, forward and aft until all bolting on the horizontal joints are tight. See Figure MI-R2.

8. Install the vertical joint bolts between the inner compressor discharge case and the compressor discharge case. Tighten and torque the bolts starting at the vertical centerline and alternately working toward the horizontal joint.

OPERATION 5 - INSTALL UPPER HALF FIRST STAGE NOZZLE SUPPORT RING

1. Clean and deburr the mating joints.

2. Apply a light coating of anti-seize compound to the horizontal joint dowels and the horizontal and vertical joint bolts.

3. Rig and lift the first-stage nozzle support ring and and lower into position on the lower half.

4. Assemble the support ring locating shim.

5. Install all horizontal and vertical bolting. Tighten and torque bolting.

OPERATION 6 - PERFORM HOT GAS PATH REASSEMBLY OPERATIONS 1 THROUGH 10



Refer to Section 4B, Hot Gas Path Inspection, for the following reassembly operations.

1. Install lower half first-stage nozzle and eccentric pin.

2. Install lower half second and third-stage nozzle segments and radial retaining pins.

3. Install lower half second and third-stage nozzle segments wheelspace thermocouples.

4. Install upper half first-stage nozzle.

5. Install upper half second and third-stage nozzle segments and radial retaining pins.

6. Recheck first-stage nozzle concentricity.

7. Recheck turbine clearances.

8. Install transition pieces.

9. Install upper half turbine casing.

10. Install wheelspace thermocouple wiring and conduit.

OPERATION 7 - INSTALL COMPRESSOR RUB RING

1. Apply a light coating of turbine oil to the compressor rub ring and assemble to the lower half inlet casing.

OPERATION 8 - INSTALL UPPER HALF COMPRESSOR INLET CASING



Check the lower half inlet casing for cleanliness and foreign matter before installing upper half inlet casing.

1. Clean and deburr the upper and lower half inlet casing horizontal joints.

2. Rig to lift the upper half casing and install guide pins.

3. Apply joint compound to the lower half horizontal joints per Section 2, solvents, sealers and cleaners, in the Service Manual.

4. Carefully lower the upper half casing to the lower half taking care to prevent casing from swinging into the rotor or adjacent flanges.

5. Apply anti-seize compound to the bolting and dowels, install bolting and dowels. Do not overlook the six inner twelve point bolts. Torque bolts alternately from the center of each flange toward each end.

6. Bolt up the inlet guide vane control ring and release the lower inlet guide vane support plate bolts.

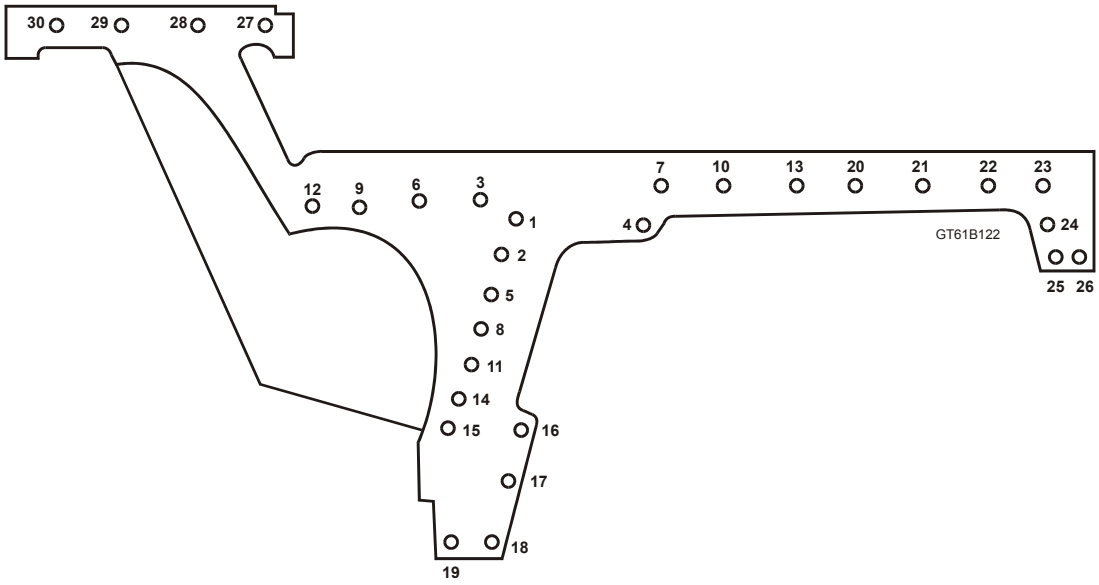


Figure MI-R2. CDC Bolt Tightening Sequence

OPERATION 9 - INSTALL FORWARD COMPRESSOR CASING

1. Clean and deburr mating joints.

2. Rig to lift the forward casing and install guide pins.

3. Check to ensure the compressor is clean and free of all foreign matter.

4. Apply joint compound on the surface of the horizontal joints only and only for a distance of one inch on both sides of the four way joints. Refer to Section 2, solvents, sealers, and cleaners, in the Service Manual.

5. Carefully lower the casing into position being careful not to swing the casing into the rotor or adjacent vertical flanges. Ensure no foreign objects enter the compressor during this operation.

6. Apply anti-seize compound to the vertical joint dowels and install the dowels between the inlet and compressor casing flanges.

7. Apply anti-seize compound to the bolting, remove the guide pins and install all bolting to horizontal and vertical flanges.

8. Torque the horizontal joint bolts starting at the mid point of the joint, working alternately left and right, forward and aft, until all bolting is tight.

9. Torque all vertical joint bolts starting at the four way joint and working alternately left and right, to top vertical centerline.

OPERATION 10 - INSTALL EXHAUST FRAME

1. Assemble new gasket to the horizontal joints of the exhaust frame.

2. Using the spreader bar rig the outer upper lugs to an overhead crane, rig to lift the exhaust frame by attaching one end of the wire rope sling to one of the lower lugs on the spreader bar and passing the other end through the lifting lugs on the exhaust frame and attach the end to the lug on the other side of the spreader bar. Repeat rigging hookup for the other side.

3. Apply a very light coating of anti-seize compound to the exhaust frame vertical flange. Lift the exhaust frame and clean and deburr the joints.

4. Position the exhaust frame over the unit and move the trolley hoist to its forward stop on the rail beam. Rig and transfer the exhaust frame to the trolley hoist.

5. Install guide pins and using the trolley hoist lower the exhaust frame into sure the exhaust cone baffle engages its mating upper half slot.

6. Apply anti-seize compound to the bolting and install all the inner vertical and horizontal joint bolts.

7. Install the outer horizontal joint bolting and the vertical bolting between the turbine casing and exhaust frame.

8. Tighten and torque all the horizontal joint bolting by starting at the mid point of the flange on both sides. Then torque alternately forward and aft both sides.

9. Tighten and torque the vertical joint bolts starting at top centerline and alternately working toward the horizontal joint.

10. Install the upper half bolts to the inner and outer diffuser upper half strut ends. Tighten and torque the bolts.



Refer to the Frame Assembly Exhaust (ML0706) drawing in the Parts Lists and Drawings Volume Service Manual and weld the bolts and nuts as called for on the outer diffuser bolts and the outer horizontal joint nuts.

11. Install the left and right side horizontal joint insulating packs. Bend up one corner of all lockplates to lock the bolts.

12. Install the upper half outer insulating pack on the aft upper vertical flange of the exhaust frame. Bend up one corner of all lockplates to lock the bolts.

13. Reconnect all pressure and temperature instrumentation to the exhaust frame and the inside of the inner barrel tunnel.

14. Using chainfall or comealong, rig to the rail beam and trolley hoist inside the exhaust transition duct. Remove the bolting from the rail beam and slide the rail and trolley out of the transition duct and rig to an overhead crane. Remove rail and hoist clear of the unit. Install all removed hardware, handtight, for safe keeping.

OPERATION 11 - REMOVE MECHANICAL SUPPORT JACKS FROM UNDERNEATH UNIT CASINGS



Mechanical support jacks are not to be removed until all upper half casing have been installed and bolted in place.

1. Remove the mechanical support jacks and dial indicators from underneath the unit casing flanges and forward inlet bell mouth flange.

2. Remove any shim packs if used. Applicable to bases that have a trans-axial "I" beam only.

3. Replace all base cutout floorplates and bolt in place.

4. Accumulate all jacks and dial indicators and return to storage.

OPERATION 12 - TAKE FINAL ROTOR POSITIONING CHECKS

1. Refer to the Hot Gas Path Inspection, EM5321, disassembly operation 3 for Rotor Positioning Checks procedures.

OPERATION 13 - PERFORM COMBUSTION INSPECTION REASSEMBLY OPERATIONS 3 THROUGH 18



Refer to EM5320, Combustion Inspection, for the following reassembly operations.

1. Install outer combustion casings and aft flow sleeves.

2. Install inner crossfire tube assemblies and check clearances.

3. Install outer crossfire tubes, adapter plates, gaskets, ring packing, and packing glands.

4. Install forward flow sleeve, inner crossfire tubes, combustion liners and spring retainers.

5. Assemble flame detectors and spark plugs.

6. Close up and bolt combustion casing hinged covers.

7. Reconnect false start drain manifold and drain lines to combustion casings.

8. Install 5th and 11th stage cooling and sealing air lines.

   NOTE Install all the 11th stage cooling and sealing air piping except the upper right side, looking with flow, section with the compressor bleed valve (33CB-2). This section will be installed after the exhaust plenum access upper panel is reinstalled.

9. Install fuel nozzles.

10. Install fuel check valves.

11. Install liquid fuel tubing lines.

12. Install water injection piping pigtails (if applicable).

13. Install atomizing air lines (including the manifold).

14. Install fuel gas piping (including the manifold) (if applicable).

15. Install steam injection piping pigtails (if applicable).

16. Install the atomizing air extraction manifold to upper half turbine casing, in addition reinstall the inlet deicing piping, if applicable, and the exhaust frame cooling piping. Install new gaskets and coat bolting with Fel Pro C-102 anti-seize compound.

OPERATION 14 - INSTALL INLET PLENUM

1. Rig to lift the inlet plenum using two lengths of timber spacing wedged forward to aft on the outer extremities to keep the plenum from distorting when being lifted.

2. Lift the upper half plenum into position onto the lower half plenum.

3. Coat bolting with Fel Pro C-102 anti-seize compound and install all vertical and horizontal joint bolts on the forward and aft walls.

4. Install new gaskets to the inlet extensions (wings) left and right sides. Lift extension wings into position and bolt in place.

5. Reconnect CO2 piping, conduit runs, wiring, switches and any other components attached to the forward and aft plenum walls.

OPERATION 15 - INSTALL INLET TRANSITION PIECE DUCT AND ELBOW

1. Check the integrity of the inlet transition duct, inlet elbow and the inlet trash screen.

2. Install new gasket on the inlet transition piece. Rig and lift the transition duct into position on the inlet plenum.

3. Coat bolting with Fel Pro C-102 anti-seize compound, install bolts and tighten.

4. Install new gaskets on the inlet duct elbow. Rig and lift the inlet elbow into position between the transition duct and silencing duct.

5. Coat bolts with anti-seize compound, install bolts and tighten.

OPERATION 16 - REASSEMBLE EXHAUST AND INLET PLENUM ACCESS PANELS

1. Spread the exhaust plenum frame using a length of 4 inch pipe or timber and a 10-ton mechanical jack sufficiently to allow clearance to install the access panel.

2. Assemble new gasket, lift and position the panel into the plenum opening. Coat bolts with Fel Pro C-102 anti-seize compound, install bolts but do not tighten.

3. Assemble new gasket to the exhaust duct transition piece forward panel. Lift and guide the panel into position.



Come-a-longs will be required to help guide and position the panel into location beneath the overhang of the transition duct.

4. Coat bolts with Fel Pro C-102 anti-seize compound. Install bolts and tighten. Remove the rigging.

5. Tighten the bolts on the exhaust plenum forward panel.

6. Reassemble the forward and aft flex seal plates, gaskets and clamp bars between the exhaust hood frame and plenum. See Fig. MI-R3.

7. Reinstall the 11th stage extraction piping and compressor bleed valve to the exhaust plenum panel. Reconnect all tubing runs and piping runs to the 11th stage piping and bleed valve.

8. Inspect the inlet plenum for cleanliness and/or foreign objects. Install new gasket on the access panel located in lower right hand corner of the plenum, looking with air flow. Coat bolts with anti-seize compound and bolt panel in place.

OPERATION 17 - RECHECK TURBINE TO LOAD GEAR AND TURBINE TO ACCESSORY GEAR ALIGNMENT

1. The turbine to load gear and turbine to accessory gear alignment should be rechecked using the same procedures detailed in disassembly Operation 1.

2. If alignment is within limits (refer to the Basic Alignment Instructions in the Service Manual), no realignment is required. Remove the alignment equipment and fixture.

3. If alignment is not within limits, realign as necessary under the direction of your GE Field Service Representative.

OPERATION 18 - INSTALL MAGNETIC PICKUP RING, ACCESSORY AND LOAD COUPLINGS

1. Reinstall the magnetic pickup ring to the front of the inlet casing and number one bearing area.

2. Lift accessory coupling. Reorient the matchmarks placed on the accessory gear to coupling, both sleeves to the coupling shaft and the turbine rotor to coupling sleeve at disassembly.

3. Add a small amount of oil to new O-rings to prevent damage to rings when installing on the hubs, install O-rings and accessory coupling bolting.

4. Torque bolts per the Basic Alignment Instruction in the Service Manual.

5. Fill the accessory coupling hubs with oil, refer to Section 2, Lubrication Guidance Chart in the Service Manual for proper oil and amount required to fill coupling. Replace fill plugs.

6. Reconnect the magnetic pickup wiring to the magnetic pickup ring.

7. The load coupling will already be in place and bolted to the turbine rotor. The load coupling will have to be reconnected to the load gear coupling flange. Refer to the Basic Alignment Instruction in the Service Manual for proper assembly and bolting procedure.

OPERATION 19 - INSTALL COUPLING GUARDS

1. Assemble the accessory gear coupling guard, short section, to the turbine end coupling guard spacer and bolt in place.

2. Assemble the long section coupling guard to the accessory gear end and over the short section. Install bolting to accessory gear end, install gasket wide washers, lockwashers and nuts to the two holes mating the two guard sections together. Tighten all bolting.

3. Assemble the bottom cover, install bolt, lockwasher and nut and tighten down.

4. Assemble load coupling guard to load gear end, install lockwashers, bolts and tighten down.

5. Assemble load coupling guard and bolt in place using lockwasher.

6. Assemble screening around aft end exhaust plenum tunnel and bolt screen to mounting blocks maintaining 0.38 inch minimum clearance around the outside diameter of load coupling guard.

OPERATION 20 - PERFORM HOT GAS PATH REASSEMBLY OPERATIONS 15 AND 16



Refer to EM5321, Hot Gas Path Inspection, for the following reassembly procedures.

1. Assemble turbine compartment roof, blowers, side panel and doors.

2. Clean up, operate and leak check unit.

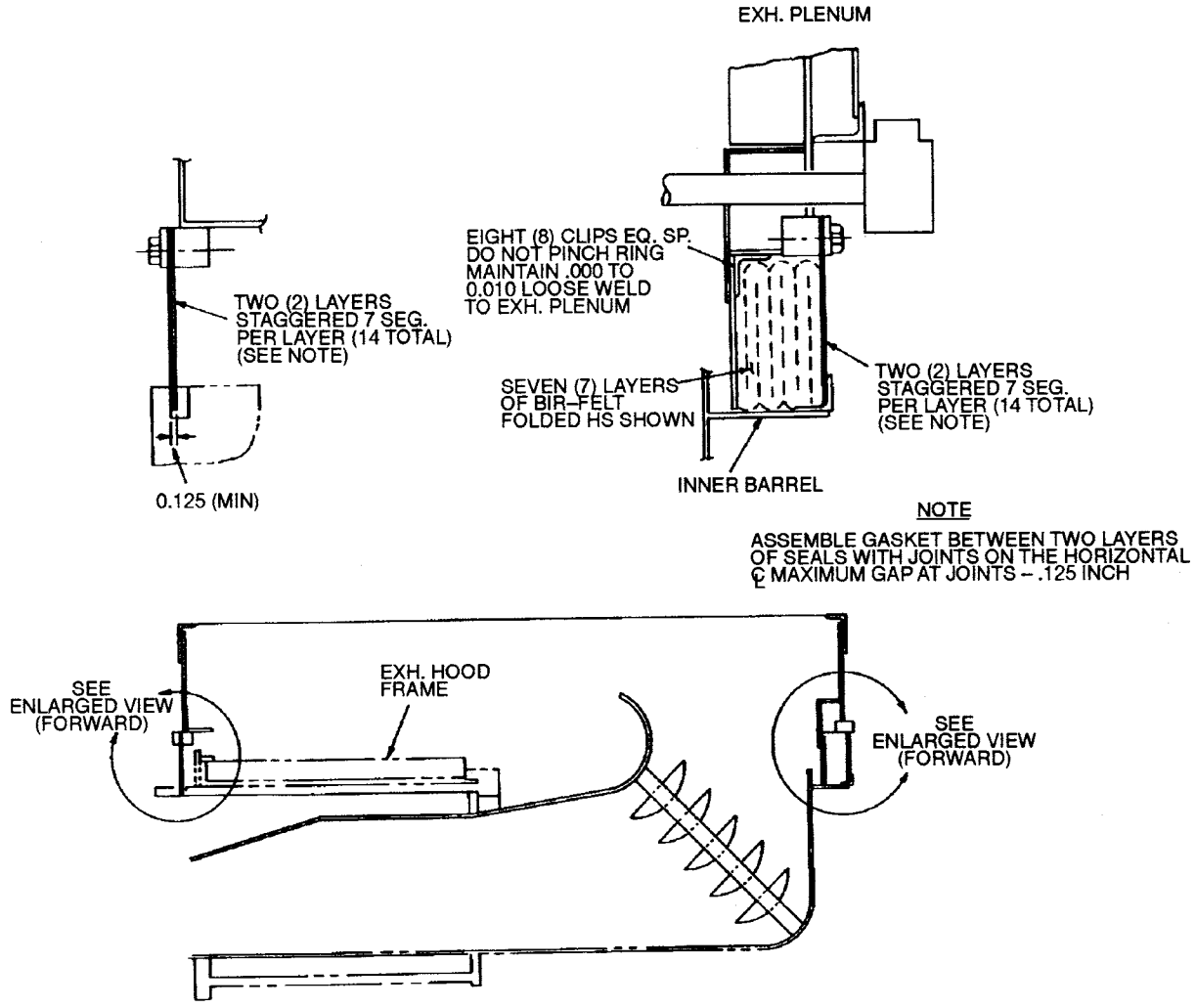


Figure MI-R3. Exhaust Plenum Seal Segments Assembly, Forward and AFT Sides

REVISION PAGE

Original Release Date
February 26, 1992

Revision Date	Revision	Revision Details	Approved By
3/31/2004	A	Added reference document EM 5202 – Gas Turbine Mechanical Support and Jacking Instructions Removed PGS form numbers and replaced with GT form numbers	MJC